Checkered Fabric AOI
Digital Twin & Quality Assistant

Project Report

Smart Factories — A.Y. 2026-2027

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Track 2 — Digital Twin, Data-Driven (Quality Monitoring)

Track 3 — RAG for Industrial Knowledge Management (Quality Assistant)

1. Introduction and Case Description

This project combines two of the four available tracks of the Smart Factories course into a single, coherent industrial case: an **Automated Optical Inspection (AOI) station** on a woven, checkered ("gingham") cotton fabric line.

Yarn is woven on a loom into the characteristic blue/yellow checked pattern, and the fabric web then passes over a finishing/inspection frame before being rolled onto the final fabric roll. A camera at the inspection frame captures images of the moving fabric surface, and each captured patch must be classified into one of three categories:

- **Circle** — a roughly circular hole / puncture in the weave
- **Line** — a linear scratch, cut, or abrasion mark across the weave
- **No defect** — a clean, regular checkered pattern

Track 2 (Digital Twin, Data-driven — Quality Monitoring) is addressed by training a ResNet18 image classifier on 150 labelled fabric images, evaluating it with stratified 5-fold cross-validation, and adding Grad-CAM explainability so an operator can verify what the model is reacting to.

Track 3 (RAG for Industrial Knowledge Management — Quality Knowledge Base) is addressed by a Quality Assistant: a retrieval-augmented system over a 12-document knowledge base of Standard Operating Procedures (SOPs), the ISO 23247 mapping, model documentation, and a textile quality standard, so that when the classifier flags a defect, the operator can immediately ask “what caused this and what should I do?” and receive a grounded, source-cited answer.

The two tracks share the same case and the same dashboard, reflecting how a real quality-monitoring digital twin and a knowledge-management assistant would be deployed together at the same workstation.

Deliverable: a 5-page Streamlit dashboard (app.py + pages/1-4), runnable locally with `streamlit run app.py`, covering EDA, the defect classifier with Grad-CAM, model performance / 5-fold CV, and the Quality Assistant.

2. ISO 23247 Reference Architecture Mapping

ISO 23247 (“Automation systems and integration — Digital twin framework for manufacturing”) defines a reference architecture of five entity types. The table below maps each entity to this project, and is what makes the prototype a *digital twin* in the ISO 23247 sense rather than a standalone ML model: it separates the physical asset, the sensing layer, the digital representation, and the user-facing application.

ISO 23247 Entity	This Project
Observable Manufacturing Element (OME)	The fabric web at the inspection frame
Device Communication Entity (DCE)	AOI camera / frame grabber (images in dataset/ stand in for a live feed)
Data Collection & Device Control	Pre-processing pipeline: resize to 224x224, normalisation, augmentation
Digital Twin Entity (Core)	ResNet18 classifier + Grad-CAM explainability module
User Entity	Streamlit dashboard — Defect Classifier + Quality Assistant pages
Cross System Entity	MES / QMS for traceability (out of scope for the prototype, discussed conceptually in Section 4.5)

TRACK 2 — DIGITAL TWIN, DATA-DRIVEN (QUALITY MONITORING)

3. Dataset and Exploratory Data Analysis

The dataset consists of **150 images** of checkered cotton fabric, perfectly balanced across the three classes (50 images each). All images are **3072x3072 px** photographs of the same blue/yellow gingham fabric sample. “Circle” and “Line” defects were created by placing a physical object (a punched hole, a metal ruler/blade) on otherwise good fabric to simulate the visual signature of those defect types — a practical approach for a small student prototype, discussed further as a limitation in Section 6.

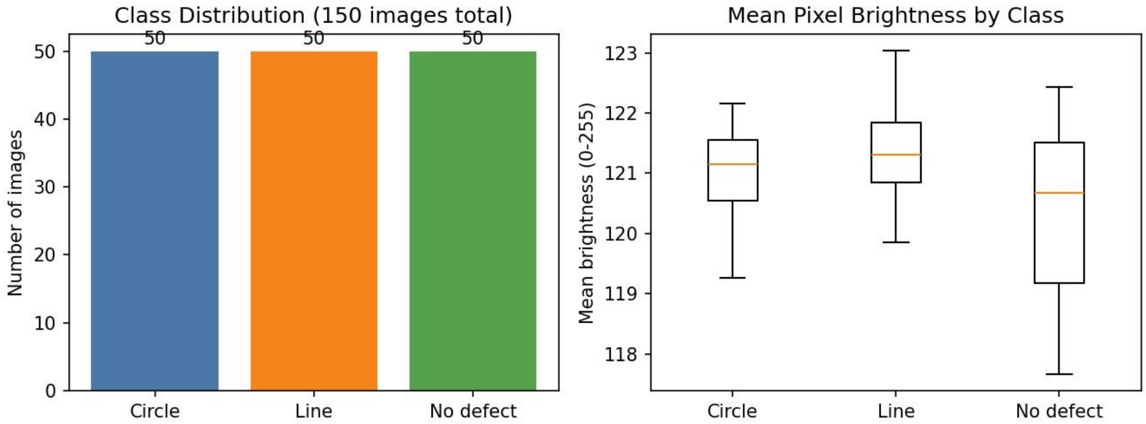


Figure 1. Left: class distribution (perfectly balanced, 50/50/50). Right: mean pixel brightness per class — no class shows a systematically different brightness, so the model cannot rely on a brightness shortcut to distinguish classes.

Data quality checks: all 150 images are valid, readable, uniquely named, and share the same resolution (3072x3072 px) and RGB colour mode — no missing values or corrupt files. All images are resized to 224x224 during pre-processing regardless of native size.

4. Model and Methodology

4.1 Architecture

The classifier is a **ResNet18** pretrained on ImageNet, with the convolutional backbone **frozen** and the final fully-connected layer replaced by a fresh 3-class linear head. Only this head is trained, for 8 epochs, with Adam (lr=1e-3) and cross-entropy loss.

Why this choice is appropriate: with only 150 images, training a full CNN from scratch (or fine-tuning the full ResNet18) would overfit badly. A frozen ImageNet backbone re-uses generic low/mid-level visual features (edges, textures, shapes) that transfer well to fabric-defect images, and the model only has to learn the final 3-class decision boundary — a much smaller and better-posed learning problem for this dataset size.

4.2 Pre-processing and Augmentation

All images are resized to 224x224 and converted to tensors. During training, random horizontal flip, random rotation (±10°), and colour jitter (brightness/contrast) are applied to reduce overfitting on the small dataset; evaluation uses a deterministic resize-only transform.

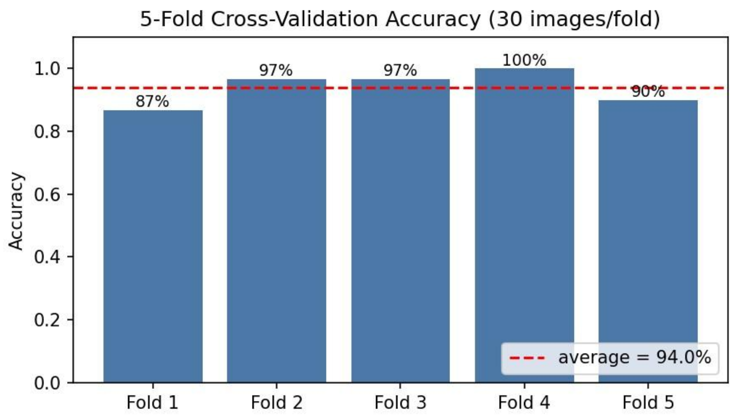
4.3 Validation Protocol

The model is evaluated with **stratified 5-fold cross-validation** (random_state=42): each fold trains a fresh 3-class head on 120 images (24 per class x 5) and tests on the remaining 30 (10 per class). This protocol gives a robust estimate of generalisation performance despite the small dataset, since every image is used for testing exactly once across the 5 folds.

5. Results

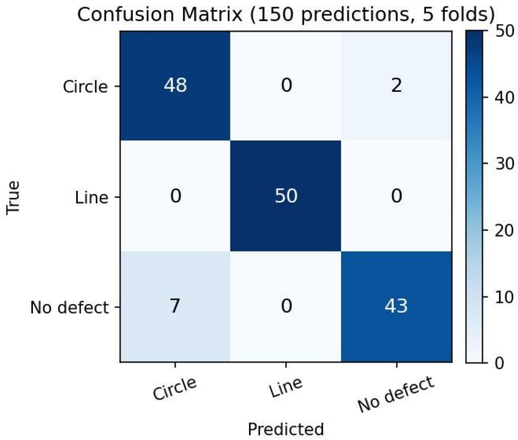
5.1 Cross-Validation Accuracy

Fold	Accuracy
Fold 1	86.7%
Fold 2	96.7%
Fold 3	96.7%
Fold 4	100.0%
Fold 5	90.0%
Average	94.0%
Std. deviation	4.9%



The model achieves an **average accuracy of 94.0%** (±4.9%) across the 5 folds, with per-fold accuracy ranging from 87% to 100%. With only 30 test images per fold, a single misclassification shifts accuracy by ~3.3 percentage points — the observed spread is expected sampling noise for this dataset size, not evidence of an unstable model, but it means the headline 94% figure should always be reported together with its standard deviation.

5.2 Confusion Matrix and Classification Report



Class	Prec.	Recall	F1	N
Circle	0.87	0.96	0.91	50
Line	1.00	1.00	1.00	50
No defect	0.96	0.86	0.91	50
macro avg	0.94	0.94	0.94	150

The largest confusion is 2 “Circle” images predicted as “No defect” and 7 “No defect” images predicted as “Circle” — i.e. confusion is concentrated between these two classes. “Line” is classified perfectly (precision = recall = 1.00). This matches the intuition that a small, faint hole can visually resemble a normal weave gap under uneven lighting, while a linear scratch is a more visually distinctive pattern.

5.3 Explainability — Grad-CAM

Grad-CAM is computed on the last residual block (layer4) of ResNet18, producing a heatmap that highlights the image regions most responsible for the predicted class. This gives the operator (User Entity) a way to sanity-check that the model is reacting to the actual defect — the hole or the scratch — rather than to lighting or staging artefacts introduced when the defect images were created.

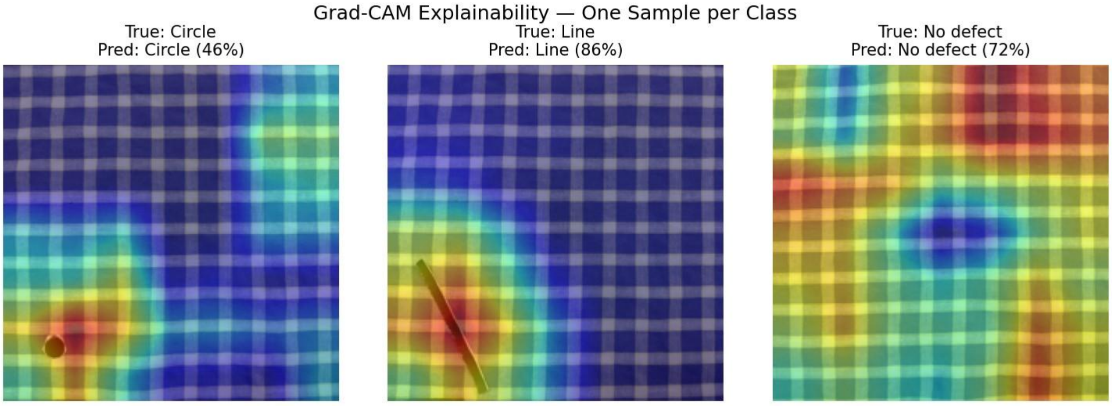


Figure 2. Grad-CAM overlays for one sample image per class (jet colormap; warm colours = higher contribution to the predicted class). The activations are broad rather than tightly localised on the defect pixels — expected for a frozen backbone where only the final linear layer is retrained (Section 6) — and the relatively low confidence on the Circle sample (46%) reflects the same limitation.

6. Critical Discussion — Track 2

Dataset size

150 images total (50/class). 5-fold CV with 8 epochs/fold gives a robust relative comparison across folds, but the absolute accuracy estimate has wide uncertainty (std = 4.9%). A production system would need hundreds to thousands of images per class, collected from the real inspection camera under production lighting.

Staged vs. real defects

“Circle” and “Line” defects were created by placing physical objects (a punched hole, a ruler/blade) on good fabric. Real weaving defects (yarn pull-outs, abrasion) may have different edge/shadow signatures — e.g. a clean punched hole lacks the frayed edges of a real yarn-pullout hole. Grad-CAM should be used to confirm the model attends to the defect region itself, not to shadows cast by the placed object.

Frozen backbone

Only the final linear layer is trained; the ResNet18 convolutional backbone keeps its ImageNet weights. This is appropriate for a 150-image dataset (avoids overfitting a much larger parameter count) but caps achievable accuracy — fine-tuning deeper layers with more data could improve discrimination between visually similar classes (Circle vs. No defect).

Single fabric pattern

The model has only seen this one blue/yellow checkered pattern. Deploying it on a different check size/colour or weave type would likely require additional labelled data and retraining.

Integration into a real system

In production, the AOI camera (DCE) would stream frames continuously; an edge device would run the same pre-processing + inference pipeline (core/model.py) at line speed, and each prediction (class, confidence, Grad-CAM region, roll ID, meterage, timestamp) would be written to the MES/QMS (Cross System Entity) for traceability and SPC charting. A confidence threshold (e.g. 0.7-0.8) would route low-confidence predictions to human review rather than auto-classifying, per the No-Defect SOP discussed in Section 7.

TRACK 3 — RAG FOR INDUSTRIAL KNOWLEDGE MANAGEMENT (QUALITY ASSISTANT)

7. Use Case and User Needs

The Quality Assistant addresses the **Quality knowledge base** objective from the RAG track: operators and quality engineers need fast, natural-language access to Standard Operating Procedures (SOPs), corrective actions, and quality standards without searching through paper manuals.

Primary user needs addressed:

- An operator who just received a “Circle” or “Line” prediction from the Defect Classifier wants toimmediately know the likely root causes and the corrective action — closing the loop from detection to action.
- A new operator wants to understand how the digital twin works (ISO 23247 mapping, modelperformance, limitations) without reading the full project documentation.
- A quality engineer wants to relate the AOI output to the mill's existing point-grading qualitystandard (4-point fabric inspection system).
- A technician wants the AOI station's maintenance/calibration checklist on demand at theinspection frame.

8. Corpus Design

The knowledge base consists of **12 documents**, written for this project as an illustrative SOP/standards corpus (no real mill manuals were available — see limitations in Section 10). Documents cover four areas: project/architecture documentation, defect-specific SOPs, a quality-standard reference, and maintenance/deployment guidance.

#	Document Title
1	Process Overview — Checkered Fabric Weaving and Inspection Line
2	ISO 23247 Mapping — Digital Twin Reference Architecture
3	What Makes This a Data-Driven Digital Twin
4	Dataset and Model — 150 Images, ResNet18, Grad-CAM
5	Model Performance — 5-Fold Cross-Validation Results
6	SOP — Circle Defect (Hole / Puncture)
7	SOP — Line Defect (Scratch / Cut Mark)
8	SOP — “No Defect” Classification and False-Negative Risk

9	Quality Standard Reference — 4-Point Fabric Inspection System
10	AOI Station Maintenance and Calibration Guide
11	Deployment and Integration in a Real Industrial System
12	Critical Discussion — Dataset and Model Limitations

Pre-processing: each document is a short (200-400 word) Markdown-formatted passage with a title. No chunking is needed at this corpus size — each document is itself a retrievable unit, which keeps citations interpretable (the operator sees a named SOP, not an anonymous text fragment).

9. System Architecture

9.1 Retrieval

Retrieval is implemented with **TF-IDF** (term frequency × inverse document frequency, computed from scratch with NumPy — no external retrieval library required, fully offline). Each query is tokenised, vectorised in the same TF-IDF space, and ranked against the 12 documents by cosine similarity; the top-3 documents are returned with their relevance scores. An optional upgrade to **sentence-transformers** (all-MiniLM-L6-v2) semantic embeddings is supported if the package is installed, for better recall on paraphrased queries.

9.2 Generation

By default (no API key), answers are composed **locally and extractively**: the system selects the sentences from the top retrieved documents with the highest token overlap with the query and assembles them under the source document's title — producing a grounded answer with zero hallucination risk and no internet dependency. If an `ANTHROPIC_API_KEY` is configured, the same retrieved context is instead passed to **Claude (claude-haiku-4-5)** with a system prompt instructing it to answer only from the provided context and to say so explicitly if the context does not contain the answer — producing a more fluent, synthesised response while remaining grounded.

9.3 User Interface

The Quality Assistant is a chat interface (Streamlit `st.chat_input`) with five example-query buttons, a persistent chat history, and a “Sources” expander under every answer showing the retrieved documents and their relevance scores. The Defect Classifier page also triggers an inline Quality Assistant query automatically whenever a defect is predicted, with a link to the full chat for follow-up questions.

10. Evaluation — 15 Representative Queries

The Quality Assistant was evaluated on 15 representative queries spanning all three required categories (the corpus does not contain hallucination-prone ambiguous content, so no query was labelled “hallucinated” — see discussion below). Full output is reproducible via `python scripts/eval_rag.py`.

#	Query	Label
1	What causes a Circle defect and how do I fix it?	Correct
2	What should I do if a Line defect is detected on the line?	Correct
3	What does a 'No defect' prediction mean and what is the false-negative risk?	Correct
4	What is the Observable Manufacturing Element (OME) in this digital twin?	Correct
5	Why is this project considered a data-driven Digital Twin rather than a simulation?	Correct
6	What model architecture is used and how was it trained?	Correct
7	What is the average accuracy of the classifier across 5-fold cross-validation?	Correct
8	Why does the per-fold accuracy vary between about 83% and 100%?	Correct
9	What is the 4-point fabric inspection system and how does it relate to this digital twin's output?	Correct
10	How should the AOI camera and inspection frame be maintained on a daily basis?	Correct
11	How would this prototype be integrated with a factory MES/QMS in a real deployment?	Correct
12	What are the main limitations of the dataset used in this project?	Correct
13	Is a Circle defect more severe than a Line defect?	Partial
14	What confidence threshold should trigger a human review of a prediction?	Partial
15	What is the current market price of cotton yarn?	No answer

Label	Count	Share
Correct	12	80%
Partial	2	13%
No answer	1	7%

Hallucinated	0	0%

Correct (12/15): queries that map cleanly onto a single document in the corpus — SOPs, ISO 23247 mapping, model performance, maintenance, and deployment integration — are all retrieved and answered correctly.

Partial (2/15): “Is a Circle defect more severe than a Line defect?” requires combining the severity notes from both defect SOPs, but the top-ranked document surfaces only one of them. “What confidence threshold should trigger human review?” touches three documents: the No-Defect SOP, which gives the only concrete value (0.8); and the Deployment/Integration and AOI Maintenance docs, which discuss threshold-based routing and alert tuning without repeating that number. Retrieval surfaces a mix of these, so the composed answer may state the general routing concept without clearly anchoring on the single concrete threshold value.

No answer (1/15): “What is the current market price of cotton yarn?” is intentionally out of scope. The system correctly retrieves only low-relevance documents and does not fabricate a price — the local extractive composer cannot hallucinate (it only assembles real sentences from the corpus), and the Claude system prompt explicitly instructs the model to say so if the context lacks the answer.

Hallucinated (0/15): the local extractive answer composer is structurally incapable of hallucinating — it can only return sentences that exist verbatim in the corpus. The Claude-backed path is grounded by an explicit “answer ONLY from the provided context” instruction, which the out-of-scope query above is designed to test.

11. Benefits, Limitations and Deployment Challenges

11.1 Benefits

- Closes the loop from detection to action: a defect prediction on the Defect Classifier pageimmediately surfaces the relevant SOP, with no manual lookup.
- Fully offline by default (TF-IDF + extractive answers) — no API costs, no internet dependency,no risk of sending proprietary process data to a third party.
- Source-cited answers: every response shows which document(s) it came from and theirrelevance score, so an operator can verify the answer against the original SOP text.
- Optional upgrade path (Claude API) for more fluent answers without changing the retrievallayer or the corpus.

11.2 Limitations

- The 12-document corpus is **illustrative**, written for this project rather than sourced from a real mill's quality manuals, maintenance procedures, or the full text of textile quality standards (e.g.
ASTM D5430).
- TF-IDF retrieval is lexical: it can miss paraphrased queries that share no vocabulary with thetarget document (mitigated by the optional semantic embedding upgrade).
- Multi-document synthesis is weak — as seen in the two “partial” evaluation results, informationsplit across documents is not always combined.
- No feedback loop: operator corrections to an assistant's answer are not currently fed back intothe corpus or retrieval ranking.

11.3 Deployment Challenges

- **Corpus maintenance:** a real deployment requires a process for keeping SOPs, standards, and maintenance procedures in the corpus up to date as they are revised — stale procedures retrieved with high confidence are worse than no answer.
- **Access control:** some documents (e.g. internal non-conformance reports) may be commercially sensitive; a real deployment needs role-based access to the corpus, not a single shared knowledge base.
- **Integration with the Cross System Entity:** in a full deployment, the assistant would also query live MES/QMS records (e.g. “what were the last 5 Circle defects on this loom?”), which requires a retrieval layer over structured data, not just static documents.
- **On-line availability:** the Claude-backed upgrade requires network access and an API key; the local extractive fallback should remain the default for a shop-floor HMI with unreliable connectivity.

12. Conclusions

This project delivered a working Streamlit dashboard implementing a data-driven Digital Twin (Track 2) and a RAG-based Quality Assistant (Track 3) for a checkered-fabric AOI station, explicitly mapped to the ISO 23247 reference architecture.

- ResNet18 (frozen backbone, 3-class head) reaches 94.0% (±4.9%) average accuracy onstratified 5-fold cross-validation over 150 images, with Grad-CAM explainability giving the operator a per-prediction view of which regions the model attended to.
- A 12-document SOP/standards knowledge base, retrieved via TF-IDF and answered eitherextractively (offline) or via Claude (optional), achieved 12/15 correct, 2/15 partial, 1/15 no-answer, and 0/15 hallucinated on the required 15-query evaluation.
- The two tracks are integrated in a single 5-page dashboard: a defect prediction automaticallytriggers the relevant SOP lookup, demonstrating the detection-to-action loop that is the practical value of combining quality monitoring with knowledge management on the factory floor.
- Key limitations — small/staged dataset, illustrative SOP corpus, single fabric pattern — aredocumented honestly in Sections 6 and 11, together with concrete recommendations (confidence-threshold human review, retraining pipeline, corpus sourced from real mill documentation) for moving from prototype to pilot.

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